

Linux Monitoring Dashboard using Grafana & Prometheus

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1. Project Overview

This project demonstrates a real-time Linux monitoring solution using Prometheus, Grafana, and Node Exporter. The goal of the project is to monitor CPU usage, memory consumption, disk utilization, and network traffic through interactive dashboards and alerts. This setup closely resembles monitoring systems used in real-world cloud, DevOps, and cybersecurity environments.

2. Technologies Used

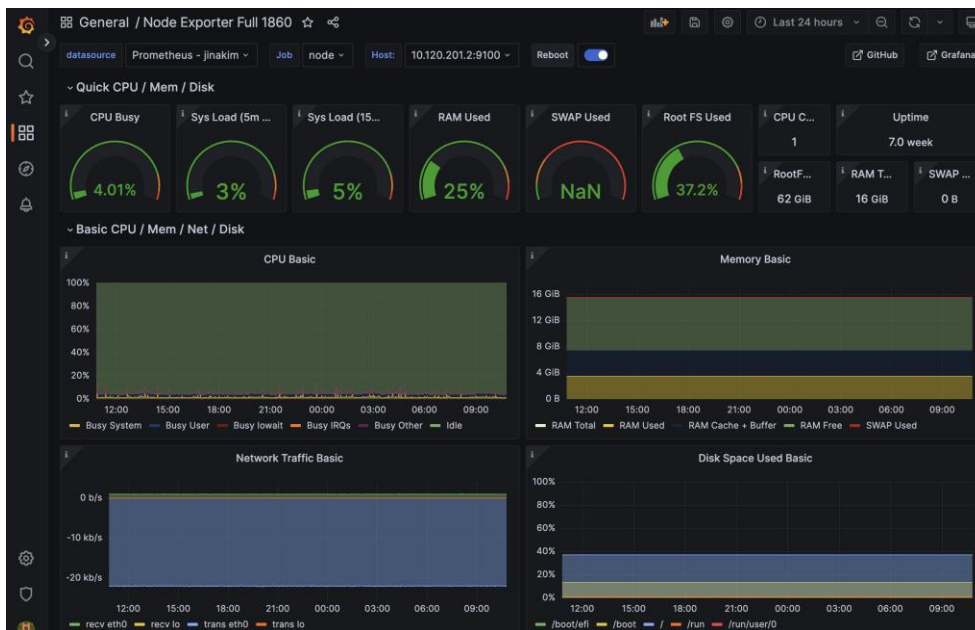
- Ubuntu Server
- Prometheus
- Grafana
- Node Exporter
- VirtualBox

3. Project Architecture

The Linux server exposes system metrics through Node Exporter. Prometheus collects these metrics at regular intervals, and Grafana visualizes them through dashboards.

4. Dashboard Screenshots

Grafana Dashboard showing CPU, RAM, and Disk utilization:



Prometheus Targets page verifying active monitoring services:

Prometheus Time Series X Shwayy

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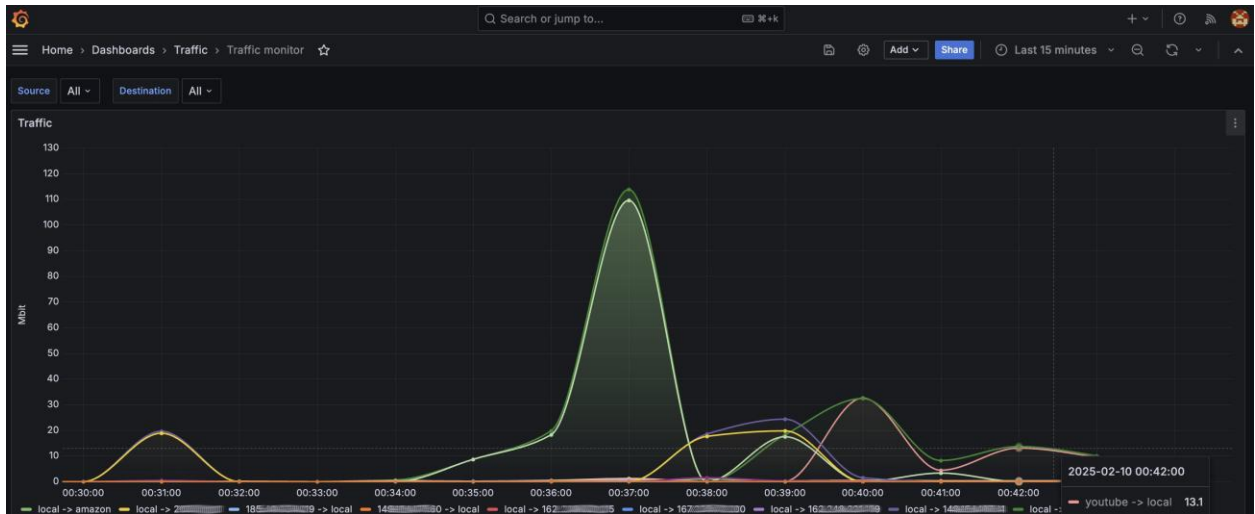
Prometheus Alerts Graph Status Help

Targets

node_exporter				
Endpoint	State	Labels	Last Scrape	Error
http://localhost:9100/metrics	UP	instance="localhost:9100"	376ms ago	

prometheus				
Endpoint	State	Labels	Last Scrape	Error
http://localhost:9090/metrics	UP	instance="localhost:9090"	4.734s ago	

Network Traffic Monitoring Dashboard:



5. Key Features

- Real-time Linux system monitoring
- CPU, RAM, and Disk usage visualization
- Network traffic monitoring
- Centralized monitoring dashboard
- Alert configuration support
- Scalable infrastructure monitoring

6. Challenges Faced

During the implementation phase, several challenges were encountered including service configuration issues, port accessibility problems, and Prometheus target connectivity errors. These issues were resolved by verifying service statuses, updating firewall rules, and validating configuration files.

7. Learning Outcomes

This project helped in understanding Linux system administration, monitoring concepts, infrastructure visualization, and real-world DevOps workflows. It also provided practical exposure to observability and server monitoring.

8. Future Improvements

- Dockerize the complete setup
- Add Loki for centralized log monitoring
- Configure Alertmanager for email notifications

- Monitor multiple Linux servers
- Integrate cybersecurity alerts

9. Conclusion

The Linux Monitoring Dashboard project successfully demonstrates how modern monitoring tools can be integrated to provide visibility into system performance and infrastructure health. This project is highly valuable for DevOps, cloud, and cybersecurity learning paths.